Mark Lights

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SUMMARY

Software Engineer with 4 years of enterprise experience at General Motors, currently completing an M.S. in Computer Science (Machine Learning) at Georgia Institute of Technology. Builds and deploys production AI systems including multi-agent LLM pipelines, RAG applications, and fine-tuned language models. Strong foundation in full-stack engineering (Java, Python, React, Spring Boot) combined with hands-on AI/ML development. All AI projects are live and self-hosted.

TECHNICAL SKILLS

AI / ML	LangGraph, LangChain, LlamaIndex, ChromaDB, HuggingFace, PyTorch, Scikit-Learn, RAG, LoRA / Fine-tuning, Prompt Engineering
Languages	Python, Java, C, JavaScript, TypeScript, HTML, CSS, SQL
Frameworks	Spring Boot, FastAPI, React.js, Next.js, Node.js, Express.js, Tuxedo
Cloud / Infra	Azure, Google Cloud, Docker, Cloudflare Tunneling, Raspberry Pi (self-hosted), Git, Gitlab
Data	Pandas, MongoDB, OracleSQL, PostgreSQL, Spring Batch
Tools	Claude Code, Aider

EXPERIENCE

Software Engineer 2 | **T-Mobile** | Dunwoody, GA | *July 2026 – Present*

- Improved billing batch job performance by reducing wall clock time from 1hr to 10 mins
- Reduced tuxedo average response time from avg 2.13s to <=500ms
- Implemented Direct Provisioning for new customers

Software Engineer | **General Motors** | Roswell, GA | *June 2022 – June 2026*

- Led production deployments across 4 enterprise applications, coordinating with QA, DevOps, and business stakeholders to deliver releases with minimal downtime in high-availability environments
- Developed and maintained REST APIs in Spring Boot with OAuth2 authentication, integrating internal microservices and downstream systems supporting business-critical workflows
- Designed Spring Batch interfaces to optimize high-volume data processing, improving throughput and concurrency across batch jobs
- Migrated enterprise application (IVH) from Spring Boot to Quarkus, containerizing services with Docker to enable cloud-native deployment on Kubernetes and reduce pod startup time and memory overhead
- Built and tuned complex SQL queries, stored procedures, and triggers, improving query performance and ensuring data integrity across transaction-heavy workflows
- Managed P1/P2 production incident queue, performing root cause analysis and implementing long-term fixes to reduce repeat incidents and improve SLA compliance
- Delivered full-stack enhancements using Spring Boot and JSP, improving user-facing workflows across business-critical systems
- Documented production deployment and rollback procedures to support release coordination and reduce recovery time
- Mentored QA engineers through a transition into software development roles

AI / ML PROJECTS

AI Interview Coach — Multi-Agent Interviewer — *May 2026 – May 2026*

- Built a multi-agent AI Interview Coach using LangGraph with parallel question generation agents, enabling dynamic, role-specific interview simulations powered by Groq LLMs
- Developed a FastAPI backend with a React frontend using Vite and CSS Modules, delivering a responsive dark UI with real-time interview feedback and session management
- Deployed the application on a self-hosted Raspberry Pi 5 with Nginx reverse proxy and Cloudflare Tunnel, using GitHub Actions CI/CD for automated production deployments via systemd
- Used Claude Code to build the UI with minimal refactoring

AI Researcher — Multi-Agent Report Generation — *Mar 2026 – Mar 2026*

- Built a three-agent LangGraph pipeline (Planner → Researcher → Writer) that takes any question and returns a fully written, source-backed research report
- Agents coordinate autonomously: the Planner decomposes queries, the Researcher performs live web search, and the Writer synthesizes findings into structured reports
- Deployed on a self-hosted Raspberry Pi 5 with Cloudflare Tunneling — publicly accessible at marks-pi.com/ai-researcher

AI Code Reviewer — Automated PR Analysis — *Mar 2026 – Mar 2026*

- Built a FastAPI + LangGraph system using LangGraph's Send API to run three parallel agents (bug detection,security analysis, performance review) concurrently against GitHub pull requests
- Agents analyze full file contents alongside diffs to eliminate false positives, then auto-post structured findings directly as GitHub PR comments
- Architected for concurrent execution across all three analysis dimensions simultaneously, minimizing total review latency
- https://marks-pi.com/ai-code-reviewer/

RAG Pipeline — Document Q&A System — *Dec 2025 – Present*

- Built a full-stack retrieval-augmented generation system allowing users to upload PDF and TXT files and query their contents via semantic search
- Implemented Python/FastAPI backend using all-MiniLM-L6-v2 embeddings and ChromaDB for vector storage and similarity search
- Built React/Express frontend and self-hosted the full stack on Raspberry Pi 5
- https://marks-pi.com/rag

Local LLaMA Voice Assistant — *May 2025 – Jun 2025*

- Deployed LLaMA 3.2 Instruct locally and built a voice-enabled interface with real-time speech-to-text and text-to-speech for hands-free conversational AI
- Optimized for low-latency on-device inference — fully private, no external API calls
- https://github.com/DevMarkLights/Ai-Assistant

Lights Finance — Portfolio Analytics App — *Nov 2023 – Nov 2025*

- Built a finance web app analyzing portfolio performance, sector diversification, dividend income, and options rate-of-return
- Integrated a custom Yahoo Finance REST API (Flask) for real-time stock and options data, and a Robinhood report parser for monthly income calculations
- Self-hosted on Raspberry Pi with Cloudflare Tunneling
- https://marks-pi.com/finance

EDUCATION

M.S. Computer Science — Machine Learning | **Georgia Institute of Technology** | *Jan 2024 – Present*

- Relevant coursework: Machine Learning (CS7641), Deep Learning (CS7643), Natural Language Processing / LLMs (CS7650), Artificial Intelligence (CS6601), ML for Trading (CS7646)
- Built a large language model from scratch using PyTorch with LSTM/RNN architecture; implemented transformers and CNNs in Deep Learning
- Trained a reinforcement learning model for stock trading strategy optimization in ML for Trading

B.S. Computer Science | **Francis Marion University** | *Aug 2015 – May 2022*

RESEARCH

Fine-Tuning Research Paper — Published research on fine-tuning large language models using parameter-efficient methods (LoRA/PEFT). https://markjarredlights.com/articles